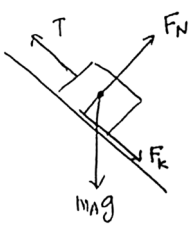


Short Qn	Part	Answer
1		Convert to <i>km</i> per litre <i>L</i> first ; $\frac{100 \text{ km}}{7.50 \text{ L}}$ Then convert to miles from <i>km</i> ; $\frac{100 \text{ km}}{7.50 \text{ L}} \times \frac{1 \text{ mile}}{1.609 \text{ km}}$ Lastly to gallons from litres ; $\frac{100 \text{ km}}{7.50 \text{ L}} \times \frac{1 \text{ US mile}}{1.609 \text{ km}} \times \frac{3.785 \text{ L}}{1 \text{ Gallon}} \cong 31.365 \text{ Gallon / US mile}$
2		$F = \Delta P \left(\frac{1.013 \times 10^5 \text{ Pa}}{1 \text{ atm}} \right) A$ Where ΔP is the pressure difference and A the area of the door . $F = (0.81 - 0.27)(1.013 \times 10^5 \text{ Pa})(1.1 \times 1.83) \cong 1.101 \times 10^5 \text{ N}$
3		Answer : B When the boat is floating, it displaces its own weight of water - the corresponding volume of water is larger than the volume of the material that makes up the boat. So when the boat sinks, it displaces less water than before so the water level decreases.
4		$\frac{P_1 V_1}{T_1} = nR = \frac{P_2 V_2}{T_2}, nR = 0.691,114, n = 0.083,127 \text{ mol}$ $V_2 = \frac{T_2}{T_1} \frac{P_1}{P_2} V_1 = \frac{7 + 273.15 \text{ K}}{20 + 273.15 \text{ K}} \frac{1.013 \times 10^5 \text{ Pa}}{7 \times 10^4 \text{ Pa}} 2.00 \text{ L}$ $\cong 2.767 \text{ L} = 2.77 \times 10^{-3} \text{ m}^3$
5		Easiest way to solve this is via area bounded by the $v(t)$ graph and the t-axis ; $S_A = \frac{1}{2} 30(8) = 120 \text{ m}$ $S_B = \frac{1}{2} (2 + 12) -25 = 2 -25 + \frac{1}{2} -25 (12 - 2) = 50 + 125 = 175$ $d_{total} = S_A + 20 + S_B = 120 + 20 + 175 = 315$
6		$x_r = +100 + 0 - 150 \cos 30.0^\circ - 200 \cos 64.0^\circ = -117.578,804 \text{ m}$ $y_r = 0 - 266 - 150 \sin 30.0^\circ + 200 \sin 64.0^\circ = -161.241,191 \text{ m}$ $r = [x_r^2 + y_r^2]^{0.5} \cong [39,823.317]^{0.5} \cong 199.6 \text{ m}$ $\tan \theta_r = \frac{y_r}{x_r} = \frac{-161.241,191}{-117.578,804} = 1.371,355$ $\theta = 180 + 53.900 \cong 233.9^\circ \text{ or } -126.1^\circ$

<u>Short Qn</u>	<u>Part</u>	<u>Answer</u>
7	a.	$\frac{d\vec{r}_{AO}}{dt} = \vec{v}(t) = (0.12 - 0.06t) \hat{i} + (-0.96 + 0.24 t^2) \hat{j} \text{ ms}^{-1} = 0$ $t = 2.0 \text{ s}$
	b.	$\frac{d\vec{v}_{AO}}{dt} = \vec{a}(t) = -0.06 \hat{i} + 0.48 t \hat{j} \quad \text{sub } t = 2.5 \text{ s into } \vec{a}(t) \text{ expression}$ $\vec{F}(2.5) = m\vec{a}(2.5) = -0.07 \hat{i} + 1.44 \hat{j} \text{ N}$
8	a.	$\Delta GPE = GPE_f - GPE_i = 14.7 \text{ J}$
	b.	$\Delta GPE = \Delta KE$ $v_f = 9.9 \text{ ms}^{-1}$
9	i.	$v_B^2 = v_A^2 + 2a\Delta y \text{ or } \frac{1}{2}mv_A^2 + mg y_A = \frac{1}{2}mv_B^2 + mgy_B$ $v_A = 3.64 \text{ ms}^{-1}$

See answers to Long Structured Questions (LQ) on the next page .

<u>Long Qn</u>	<u>Part</u>	<u>Answer</u>
1	a.	$600 \times 50 = P \times 30$ $P = 1000 \text{ MW}$
	b.	$m = \frac{Q}{c \cdot \Delta T} = \frac{8,000}{4,186 \times 15.0} \cong 0.1274 \text{ kg s}^{-1} = 127.4 \text{ g s}^{-1}$
	c.	Liquid cooling is preferred to air cooling because of Higher heat capacity. Liquids like water have a much higher specific heat capacity than air (water: $4186 \text{ J/kg}^\circ\text{C}$; air: $\sim 1000 \text{ J/kg}^\circ\text{C}$). This means water can absorb and carry away significantly more heat energy per unit mass and temperature change , making it far more effective at cooling high-powered components.
2	ai.	$v_x = 34.8 \cos 33.5^\circ \cong 29.02 \text{ ms}^{-1}$ $x = v_x t = 138.4 \text{ m}$
	aii.	$v_x = v_0 \cos \theta_0 = 29.0 \text{ ms}^{-1}$ $\theta_0 = 43.5^\circ$
	bi.	49.02°
	bii.	$h = v_{iy} t + \frac{1}{2}(-g) t^2 \Rightarrow t = 5.41 \text{ s}$ $x = v_0 \cdot \cos \theta_{opt} \cdot t = 141.9 \text{ m}$ $d = 141.9 - 138.4 = 3.5 \text{ m}$
3	ai.	 Free-body diagram , 1 mark for 1 force F_{Kf} – kinetic friction force
	aii.	For block A ; $F_N = m_A g \cos \theta$ $T - \mu_K m_A g \cos \theta - m_A g \sin \theta = m_A a$ For block B ; $m_B g - T = m_B a \quad (\text{note this is before rotational dynamics , Tension taken to be the same})$ $m_B = \frac{a + g(\mu_K \cos \theta - \sin \theta)}{g - a} m_A \cong 9.62 \text{ kg}$
	b.	$F_N = F \sin \theta + mg$ $F \cos \theta - F_{Kf} = m a$ $a = \frac{F \cos \theta - \mu_K (F \sin \theta + mg)}{m} \cong 3.04 \text{ ms}^{-2}$ 